Code No: 115AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What are attributed grammars? Give examples.	[2]
b)	Discuss about virtual machines.	[3]
c)	What is meant by short circuit evaluation of an expression?	[2]
d)	Explain about guarded commands.	[3]
e)	What does it mean by Scope and lifetime of variable?	[2]
f)	What are called recursive subprograms? Explain.	[3]
g)	State Inference process of prolog.	[2]
h)	What are the applications of logic programming?	[3]
i)	What are the applications of functional programming languages?	[2]
j)	What is procedural abstraction? Give example.	[3]

PART - B

(50 Marks)

2.a) What are the important factors influencing the writability of a language? Discuss.
b) Construct the parse tree for the simple statement: A := B * (A + C). [5+5]

OR

3.a) Explain about various programming domains and their associated languages.
b) Discuss the significance of logic programming in detail. [5+5]

4.a) What are the problems associated with Unconditional Branching? Explain with examples.
b) Discuss about various design issues of character string types. [5+5]

OR

5.a) What are dangling pointers and lost heap-dynamic variables? How are they created? Explain.
b) Explain the different types of Union with examples. [5+5]

6.a) With examples, explain the problem of passing multi-dimensioned arrays as parameters.
b) Discuss in detail about the significance of Co-Routines with examples. [5+5]

OR

7.a) Explain type checking technique in parameter passing. Give suitable example.
b) Describe about the static and dynamic scope of variables with an example. [5+5]

8.a) What are the basic elements of Prolog? Discuss.
b) How to handle exceptions in Ada? Explain. [5+5]

OR

9.a) List and explain different types of propositions present in logic programming.
b) Demonstrate the concept of abstraction and encapsulation with suitable examples. [5+5]

10.a) Make a comparison between imperative and functional languages.
b) Write a LISP function fib(n) that computes nth Fibonacci number. [5+5]

OR

11.a) Write a detailed note on Scripting languages.
b) How the storage and control is handled in Python? Explain. [5+5]

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